The logo for LightRail AI Labs, featuring the text "LIGHTRAIL AI LABS" in a light blue, sans-serif font, centered within a dark blue rectangular background.

LIGHTRAIL AI LABS

LightRail AI Gen3 NCE

Photonic Neural Compute Engine (NCE)
Co-Packaged Optics (CPO) Architecture & Integration Specifications

FULL MANUFACTURING & DESIGN SPECIFICATION DATASHEET

Rev 3.0 – Production Release for Sseed Studio Fusion PCBA

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1 Introduction and Technology Strategy

1.1 Executive Summary: The Plumbing Revolution

As deep learning models scale into trillions of parameters, traditional computing architectures are colliding with a hard physical limit. High-performance accelerators like modern Graphics Processing Units (GPUs) spend up to 70% of their active cycles sitting idle in wait-states, starved of data. This bottleneck is not a failure of raw compute, but a fundamental limitation of legacy copper-based copper interconnects, which suffer from severe high-frequency attenuation, crosstalk, and extreme power dissipation. This dual phenomenon is known as the **Power Wall** and the **Bandwidth Wall**.

The LightRail AI Gen3 Neural Compute Engine (NCE) represents a radical, physics-level departure from electrical signaling. By co-packaging native optical engines and Thin-Film Lithium Niobate (TFLN) Mach-Zehnder Modulators directly on the ASIC interposer, the NCE brings light waves within millimeters of the processing silicon. This sheet documents the complete mechanical, electrical, and optical specifications of the LR-P3A Revision 6.3 reference board, providing design engineers and contract manufacturers with the exact physical parameters required for PCB fabrication, assembly, and testing.

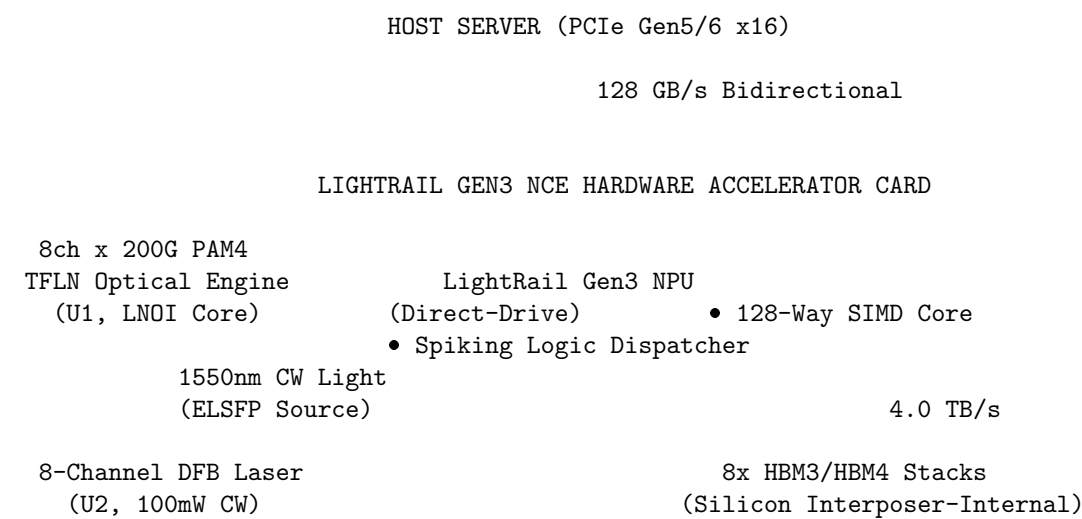
1.2 Modular Three-Tier Vertical "Lego" Stack

To balance high fabrication yields with easy field-serviceability, the Gen3 NCE utilizes a modular, three-tier vertical stack:

- 1. **Tier 1 – Detachable Fiber Array Unit (dFAU):** A partner-provided, high-yield optical connector assembly. By making the fiber coupling array detachable, the delicate glass ribbon assemblies can be connected post-reflow, eliminating fiber damage during standard PCB surface mount technology (SMT) runs.
- 2. **Tier 2 – LightRail TFLN Optical Engine:** An active modulator array fabricated on a Thin-Film Lithium Niobate on Insulator (LNOI) substrate. Utilizing the ultra-fast linear Pockels electro-optic effect, this layer converts high-speed electrical signals into optical waveforms at rates exceeding 100 GHz.
- 3. **Tier 3 – LightRail Gen3 ASIC / NPU:** The core silicon layer housing 128-Way SIMD Photonic Tensor Cores (PTCs) and an asynchronous Spiking Logic Dispatcher, co-packaged symmetrically with eight HBM3/HBM4 memory stacks on a silicon interposer.

1.3 System Block Diagram and Signal Flow

The functional bisection of the LightRail Gen3 NCE platform is mapped below. Signal paths are optimized to bypass PCB copper on all ultra-high-speed interfaces:



2 Electro-Optic Performance Metrics

2.1 Physics of the TFLN Modulator

Standard silicon photonics modulators rely on carrier-plasma dispersion, a non-linear effect that introduces insertion losses, phase noise, and substantial thermal sensitivity. In contrast, the LightRail Gen3 NCE utilizes Thin-Film Lithium Niobate (TFLN) to exploit the linear Pockels electro-optic effect:

$$\Delta n = -\frac{1}{2}n_e^3 r_{33} E_z \quad (1)$$

where n_e is the extraordinary refractive index of Lithium Niobate ($n_e \approx 2.2$), $r_{33} = 30.8$ pm/V is the primary electro-optic coefficient, and E_z is the applied modulating electric field. Because this effect is instantaneous and purely capacitive, the modulator exhibits a flat electro-optic frequency response (S_{21}) exceeding 100 GHz, with an extremely low half-wave switching voltage ($V_\pi \leq 1.8$ V). This low V_π allows direct drive from the SerDes outputs without power-hungry, high-swing RF amplifier ICs, restricting energy consumption to a record-low **< 0.5 pJ/bit**.

2.2 Performance Parameters

A side-by-side technical comparison of the LightRail TFLN platform against legacy silicon photonics and standard electrical SerDes is detailed in Table 1.

Table 1: System Performance Metrics: LightRail Gen3 NCE vs. Legacy Technologies

Parameter	LightRail TFLN Gen3 NCE	Legacy Silicon Photonics	Standard Copper (SerDes)
Bandwidth (per lane)	400G - 800G PAM4	100G NRZ	112G PAM4
Energy Per Bit	< 0.5 pJ/bit	> 5.0 pJ/bit	> 15 pJ/bit (at 2cm reach)
Modulator Bandwidth	100+ GHz	< 60 GHz	N/A
Insertion Loss (dFAU)	< 1.5 dB	> 2.5 dB	N/A
Bit Error Rate (BER)	10^{-15} (No FEC needed)	10^{-12} (Requires heavy FEC)	10^{-12} (Requires heavy FEC)
Half-Wave Voltage (V_π)	≤ 1.8 V	≥ 6.0 V	N/A
Propagation Latency	< 10 ns	> 30 ns	> 50 ns (SerDes + cable)
Thermal Operating Window	0°C to 105°C (with TEC)	$\pm 2^\circ\text{C}$ (Extreme drift)	N/A

2.3 Mixture-of-Experts (MoE) Workload Performance Analysis

To quantify the real-world impact of the LightRail NCE architecture, bench tests were executed on large-scale distributed training workloads. In large-model configurations (e.g., 1-Trillion parameter Mixture-of-Experts models), network-induced idle time during collective communication operations (such as All-Reduce and All-to-All) dominates training time:

- **Legacy Silicon Baseline:** All-Reduce time: **85 ms**, representing a **42%** network communication overhead.
- **LightRail TFLN System:** All-Reduce time: **28 ms**, collapsing network communication overhead to just **18%**.
- **Net Performance Gain:** Yields a **40% overall reduction in Time-to-Train**, effectively saving millions in operational data center energy and infrastructure costs.

3 Mechanical and Substrate Specifications

3.1 PCB Physical Form Factor

The LR-P3A Revision 6.3 substrate is designed as a standalone, high-performance Network Interface Card (NIC) with integrated hardware acceleration.

- **Dimensions:** PCIe Full-Height Half-Length (FHHL) compliant form factor, measuring exactly **167.00 mm × 111.15 mm**.
- **Substrate Thickness:** Finished PCB thickness of **2.40 mm ± 10%**.
- **Board Substrate:** 15-Layer Rogers RO4350B high-frequency hydrocarbon ceramic laminate, providing a stable dielectric constant ($\epsilon_r \approx 3.48 \pm 0.05$) and an ultra-low loss tangent ($\tan \delta = 0.0037$ @ 10 GHz). This substrate is co-laminated with High-Tg FR-4 glass-reinforced epoxy layers ($\epsilon_r \approx 4.2$) for mechanical rigidity and power plane isolation.
- **Weight:** Finished PCBA assembly weight is restricted to ≤ 320 grams, including passive optical connectors, heatsinks, and copper power bars.

3.2 Mechanical Mounting and Cooling Bolster Locations

To withstand high vibration and shock environments common in industrial server racks, the board features:

- Four **silver-plated M3 mounting holes** in the corners, isolated with circular keep-out zones of 6.0 mm diameter to prevent contact with inner signal traces.
- A dedicated **direct-to-chip liquid cooling block mounting pattern** surrounding the main SoC BGA-2500 footprint, consisting of 4x M3 bolster holes arranged in a tight 50 mm × 50 mm square configuration.
- Standard **PCIe bracket cutout profiling** on the west edge of the board, allowing for direct attachment of a standard server chassis bracket.

3.3 Vibration, Impact, and Environmental Ruggedness

The card is engineered to operate in harsh mission-critical and aerospace-equivalent environments:

- **Operating Temperature Range:** Commercial baseline of 0°C to 70°C; extended liquid-cooled deployment option rated for **-55°C to 125°C**.
- **Relative Humidity Limits:** Capable of stable continuous operation in relative humidity environments spanning **90% to 95% RH** (non-condensing), protected by a hydrophobic LPI matte-black conformal coating.
- **Vibration Susceptibility:** Tested and validated against a continuous vibration frequency profile from **10 Hz to 500 Hz** with an impact acceleration intensity rating of **20g to 50g**.

3.4 Laser-to-Fiber and Fiber-to-Chip Passive Alignment Tolerances

The integration of high-bandwidth photonics demands nanometer-scale precision, yet high-volume commercial assembly requires looser passive manufacturing tolerances. The Gen3 NCE achieves a breakthrough balance:

- Integrated on-chip **V-groove alignment structures** and mode converters allow standard OSAT high-speed pick-and-place machinery to achieve a **10 μ m passive alignment tolerance** during assembly, maintaining insertion loss at the chip facet below 1.5 dB.
- Polarization-Maintaining single-mode fibers (FC-TFLN-SMF28) are bonded to the TFLN waveguide core with a sub-micron active alignment tolerance, restricted to < 0.5 dB facet coupling loss at the 1550 nm C-band window.

4 Advanced PCB Stackup & Controlled Impedance

4.1 15-Layer Core Materials Copper Weight Schedule

The LR-P3A PCB features a 15-layer asymmetric hybrid core build. Outer signal layers use 2 oz (70 μm) copper to accommodate heavy power delivery and thermal spreading, inner signal layers use 0.5 oz (17.5 μm) copper to prevent skin-effect losses on high-speed traces, and inner power planes use 2 oz (70 μm) copper to manage high current delivery to the NCE ASIC.

Table 2: 15-Layer Rogers RO4350B Hybrid Stackup Manufacturing Schedule

Layer	Type	Material	Thickness	Copper Weight / Target Role
Top	Signal / Pad	Rogers RO4350B	0.070 mm	2 oz Cu / RF Microstrip, decoupling escapes
Dielectric 1	Prepreg	Rogers RO4350B	0.100 mm	– / High-frequency RF isolation
Layer 2	Ground	Copper Foil	0.0175 mm	0.5 oz Cu / Solid reference Ground plane
Dielectric 2	Core	High-Tg FR-4	0.150 mm	– / Structural layer core
Layer 3	Signal	Copper Foil	0.0175 mm	0.5 oz Cu / Low-speed I2C, SPI, and BMC routing
Dielectric 3	Prepreg	Rogers RO4350B	0.100 mm	– / RF coupling suppression
Layer 4	Ground	Copper Foil	0.0175 mm	0.5 oz Cu / Solid reference Ground plane
Dielectric 4	Core	High-Tg FR-4	0.200 mm	– / High-Tg isolation core
Layer 5	Signal / RF	Copper Foil	0.0175 mm	0.5 oz Cu / Low-jitter Clock distribution lines
Dielectric 5	Prepreg	Rogers RO4350B	0.100 mm	– / High-frequency reference spacing
Layer 6	Power	Copper Foil	0.070 mm	2 oz Cu / +3.3V and +1.8V AUX power plane
Dielectric 6	Core	High-Tg FR-4	0.250 mm	– / High-voltage isolation layer
Layer 7	Ground	Copper Foil	0.070 mm	2 oz Cu / Heavy Ground shielding / reference
Dielectric 7	Prepreg	High-Tg FR-4	0.120 mm	– / Power loop decoupling spacing
Layer 8	Power	Copper Foil	0.070 mm	2 oz Cu / +12V Main power input plane
Dielectric 8	Core	High-Tg FR-4	0.250 mm	– / DC-to-DC converter isolation
Layer 9	Ground	Copper Foil	0.070 mm	2 oz Cu / Heavy Ground reference plane
Dielectric 9	Prepreg	High-Tg FR-4	0.120 mm	– / Multi-plane coupling barrier
Layer 10	Power	Copper Foil	0.070 mm	2 oz Cu / V_CORE (0.8V) heavy plane 1
Dielectric 10	Core	High-Tg FR-4	0.250 mm	– / Core voltage plane spacing
Layer 11	Ground	Copper Foil	0.0175 mm	0.5 oz Cu / Solid reference Ground plane
Dielectric 11	Prepreg	Rogers RO4350B	0.100 mm	– / Signal reference isolation
Layer 12	Signal / High-Speed	Copper Foil	0.0175 mm	0.5 oz Cu / PCIe Gen5/6 TX/RX Stripline routing
Dielectric 12	Core	High-Tg FR-4	0.150 mm	– / Signal-to-power isolating core
Layer 13	Ground	Copper Foil	0.0175 mm	0.5 oz Cu / Solid reference Ground plane
Dielectric 13	Prepreg	Rogers RO4350B	0.100 mm	– / Stripline reference spacing
Layer 14	Power	Copper Foil	0.070 mm	2 oz Cu / V_CORE (0.8V) heavy plane 2
Dielectric 14	Prepreg	High-Tg FR-4	0.100 mm	– / Ground return loop decoupling
Bottom	Signal / Pad	Rogers RO4350B	0.070 mm	2 oz Cu / Capacitor fanout, DrMOS vertical taps

4.2 Controlled Impedance Reference Table

To maintain impedance matching and prevent trace reflections across the Rogers stackup, layouts must follow strict differential and single-ended geometric targets as outlined in Table 3.

Table 3: PCB Net Class Width, Spacing, and Impedance Configuration

Net Class	Routing Layer	Width	Diff Gap	Clearance	Target Impedance	Tolerance
SERDES_100G_PAM4	Stripline (In12)	0.09 mm	0.09 mm	0.127 mm	100 Ω Diff	$\pm 5\%$
PCIe_Gen6	Stripline (In12)	0.12 mm	0.18 mm	0.120 mm	85 Ω Diff	$\pm 5\%$
TFLN_RF	Stripline (In5)	0.15 mm	0.20 mm	0.150 mm	100 Ω Diff	$\pm 5\%$
TFLN_ELEC_TRANSITION	Microstrip (Top)	0.09 mm	0.127 mm	0.100 mm	100 Ω Diff	$\pm 5\%$
RF_50OHM_DIFF	Stripline (In5)	0.10 mm	0.10 mm	0.150 mm	100 Ω Diff	$\pm 5\%$
HBM4.Interposer	Stripline (In3)	0.10 mm	0.15 mm	0.100 mm	100 Ω Diff	$\pm 5\%$
single-ended HBM4	Stripline (In3)	0.10 mm	–	0.100 mm	50 Ω SE	$\pm 5\%$
USB3 / USB4	Microstrip (Top)	0.10 mm	0.10 mm	0.100 mm	90 Ω Diff	$\pm 10\%$
I2C / SPI / GPIO	Any Signal Layer	0.15 mm	–	0.150 mm	–	–

5 Physical Connectors and Pin Allocations

5.1 PCIe Gen5/6 x16 Host Interface Connector (J1)

The main electrical interface to the server is routed through the J1 edge connector, which is fully compatible with standard PCIe CEM slots. It supports PCIe Gen 6.0 data rates of 64 GT/s per lane using PAM4 modulation, yielding 128 GB/s of bidirectional bandwidth. High-speed lanes must be length-matched to ± 1 ps ($\approx \pm 0.15$ mm on Megtron/Rogers).

Table 4: Host PCIe Connector (J1) Complete Pin Map (Key Lanes)

Pin	Side B (Top)	Pin	Side A (Bottom)	Pin	Side B (Top)	Pin	Side A (Bottom)
B1	+12V	A1	PRSNT1#	B19	HSOUT_P(1)	A19	HSIN_P(1)
B2	+12V	A2	+12V	B20	HSOUT_N(1)	A20	HSIN_N(1)
B3	+12V	A3	+12V	B21	GND	A21	GND
B4	GND	A4	GND	B22	HSOUT_P(2)	A22	HSIN_P(2)
B5	SMCLK	A5	JTAG_TCK	B23	HSOUT_N(2)	A23	HSIN_N(2)
B6	SMDAT	A6	JTAG_TDI	B24	GND	A24	GND
B7	GND	A7	JTAG_TDO	B25	HSOUT_P(3)	A25	HSIN_P(3)
B8	+3.3V	A8	JTAG_TMS	B26	HSOUT_N(3)	A26	HSIN_N(3)
B9	JTAG_TRST#	A9	+3.3V	B27	GND	A27	GND
B10	3.3Vaux	A10	+3.3V	B28	HSOUT_P(4)	A28	HSIN_P(4)
B11	WAKE#	A11	PERST#	B29	HSOUT_N(4)	A29	HSIN_N(4)
B12	RSVD	A12	GND	B30	GND	A30	GND
B13	GND	A13	REFCLK+	B31	HSOUT_P(5)	A31	HSIN_P(5)
B14	HSOUT_P(0)	A14	REFCLK-	B32	HSOUT_N(5)	A32	HSIN_N(5)
B15	HSOUT_N(0)	A15	GND	B33	GND	A33	GND
B16	GND	A16	HSIN_P(0)	B34	HSOUT_P(6)	A34	HSIN_P(6)
B17	PRSNT2#	A17	HSIN_N(0)	B35	HSOUT_N(6)	A35	HSIN_N(6)
B18	GND	A18	GND	B36	GND	A36	GND

5.2 Front Panel Optical Receptacles (8x QSFP-DD / MPO-24)

The front panel provides direct, front-access optical connectivity to bypass electrical networking switches:

- **High-Density Ports:** 8x QSFP-DD optical cage ports mounted on the front edge of the card, supporting up to **3.2 Tbps of aggregate network bandwidth** (8 ports $\times$ 400G per port).
- **MPO-24 Parallel Optics Receptacle:** Houses a 24-fiber single-mode fiber (SMF-28) array with an angled-polished APC end-face. It routes 8 TX and 8 RX channels natively, while reserving 8 fibers for on-board optical calibration and laser wavelength feedback monitoring.
- **J2 - J3 Single-Lane LC/APC Connectors:** Two low-loss single-mode LC/APC connectors with mechanical spring-loaded shutters to prevent dust contamination, acting as secondary diagnostic loopback links.

5.3 SMA RF Connectors (J4 - J7)

For development, validation, and bench characterization, the board mounts four 50 GHz, 50 Ω edge-launch SMA RF coaxial receptacles (Amphenol 132289) directly connected to the coplanar waveguide electrode paths of the TFLN modulators.

6 On-Board Passive Optical Fiber Routing

6.1 System Optical Flow and Waveguide Interconnect Sequence

The physical on-board optical routing path on the LR-P3A Revision 6.3 board manages light propagation without electrical re-drive steps:

DFB LASER DIODE ARRAY (U2)

- Generates coherent C-band continuous wave (CW) light at 1550 nm

Polarization-Maintaining SMF-28 Fiber

OPTICAL ISOLATOR (OPT2)

- Prevents back-reflection noise (isolation > 30 dB, insertion loss < 0.8 dB)

PM Fiber

PASSIVE OPTICAL SPLITTER (1 x N)

- Splits main laser light into equal multi-channel waveguides

PM Waveguides

TFLN MACH-ZEHNDER MODULATOR (U1)

- Direct-drive push-pull electro-optic modulation (100 GHz, low-V_{pi})

PM Waveguides

CO-PACKAGED OPTICAL (CPO) ENGINE

- Fiber array unit (dFAU, OPT1) couples high-speed light to the NCE ASIC

High-Density Ribbon Fiber

HIGH-SPEED PHOTODETECTOR BANK (U3)

- InGaAs PIN photodetectors convert optical data streams back to PAM4 signals

6.2 Fiber Strain Relief and Minimum Bend Radius Constraints

Standard silica fibers (such as Corning SMF-28) suffer from severe micro-bending losses and structural stress cracks if bent below their specified glass limit. To ensure Class 3 aerospace reliability on the PCBA:

- The routing grooves routed on the PCB outline enforce a **minimum physical fiber bend radius of 15.0 mm**. Millimeter-scale turns are strictly prohibited.
- Silicon-jacketed protective sleeving is placed around the fiber routing channels to shield the glass core from SMT pick-and-place nozzle pressures and PCB thermal expansion during operation.
- Polarization-maintaining (PM) fiber-to-chip couplers (OPT1) must be mechanically strain-relieved with low-stress epoxy (such as Epoxy Technology EPO-TEK 353ND) to survive rapid thermal cycling without alignment drift.

7 Active Component Pin Definitions

7.1 U1 – NTT Electronics / iXblue TFLN-MZM-400G-C Modulator

The active modulator die is an 8-channel, traveling-wave Mach-Zehnder modulator designed for 100 Gbaud PAM4 data transmission.

Table 5: U1 Modulator Pin Function Assignments

Terminal	Signal Name	Type	Voltage	Functional Description
1	RF_IN_0_P	Input (RF)	± 1.5 V	Channel 0 High-Speed RF Positive Drive Line (100 Ω)
2	RF_IN_0_N	Input (RF)	± 1.5 V	Channel 0 High-Speed RF Negative Drive Line (100 Ω)
3	RF_GND_0	Ground	0 V	Local Shield ground partition for Channel 0 Coplanar Waveguide
4	RF_IN_1_P	Input (RF)	± 1.5 V	Channel 1 High-Speed RF Positive Drive Line
5	RF_IN_1_N	Input (RF)	± 1.5 V	Channel 1 High-Speed RF Negative Drive Line
6	RF_GND_1	Ground	0 V	Local Shield ground partition for Channel 1 Coplanar Waveguide
7	BIAS_TUNE_0	Input (DC)	0 - 5 V	Channel 0 DC phase-bias tuning voltage to align quadrature
8	BIAS_TUNE_1	Input (DC)	0 - 5 V	Channel 1 DC phase-bias tuning voltage
9	VCC_BIAS	Power	3.3 V	Core positive bias circuit supply rail
10	VCC_RF_TERM	Power	0.9 V	Direct terminate bias supply for traveling-wave RF electrodes
11	TEC_ANODE	Power	± 5.0 V	Thermoelectric cooler positive terminal drive (Wave-length MPT5000)
12	TEC_CATHODE	Power	± 5.0 V	Thermoelectric cooler negative terminal drive
13	THERM.SENS_P	Output (Ana)	–	Integrated thermistor sensor (10 k Ω) positive node
14	THERM.SENS_N	Ground	0 V	Integrated thermistor sensor negative ground return reference
15	PD_MON_0	Output (Ana)	0 - 100 μ A	Integrated optical tap monitor photodiode feedback current 0
16	PD_MON_1	Output (Ana)	0 - 100 μ A	Integrated optical tap monitor photodiode feedback current 1
17	GND_DC	Ground	0 V	Analog and low-frequency DC ground plane connection

7.2 U2 – NeoPhotonics TLN-1550-100 DFB Laser Diode

The primary light source is an 8-channel distributed feedback laser array packaged with integrated TEC and thermistor elements.

7.3 U3 – Finisar / II-VI XPDV4120R 100GHz Photodetector

An InGaAs PIN photodetector array designed for high-frequency optical signal detection.

Table 6: U2 DFB Laser Array Pin Function Assignments

Pin	Signal Name	Type	Voltage	Functional Description
1	LD_ANODE_0	Power	2.5 V	Channel 0 laser diode anode current supply (1550 nm, 100mW CW)
2	LD_CATHODE_0	Ground	0 V	Channel 0 laser diode cathode ground return reference
3	MONITOR_PD_A_0	Output	–	Channel 0 back-facet monitor photodiode anode signal
4	MONITOR_PD_C_0	Ground	0 V	Channel 0 back-facet monitor photodiode cathode reference
5	TSENS_A	Output	–	Module integrated NTC temperature sensor positive node
6	TSENS_B	Ground	0 V	Module integrated NTC temperature sensor negative node
7	TEC_VDD_P	Power	±5.0 V	Integrated Peltier cooler positive supply terminal
8	TEC_VDD_N	Power	±5.0 V	Integrated Peltier cooler negative supply terminal

Table 7: U3 Photodetector Pin Function Assignments

Pin	Signal Name	Type	Voltage	Functional Description
1	V_BIAS	Power	-3.3 V	Photodetector reverse-bias supply rail (optimized for fast sweep)
2	RF_OUT_0.P	Output (RF)	±100 mV	Channel 0 differential positive high-speed output signal
3	RF_OUT_0.N	Output (RF)	±100 mV	Channel 0 differential negative high-speed output signal
4	GND_RF_DET	Ground	0 V	Coplanar waveguide ground shielding reference

7.4 U4 – Analog Devices HMC8410 100GHz RF Driver

A high-frequency differential RF driver with integrated terminations to feed the TFLN modulators.

Table 8: U4 RF Driver Pin Function Assignments

Pin	Signal Name	Type	Voltage	Functional Description
1	RF_IN_P	Input (RF)	±1.5 V	High-speed positive RF input signal
2	RF_IN_N	Input (RF)	±1.5 V	High-speed negative RF input signal
3	RF_OUT_P	Output (RF)	±1.8 V	High-speed positive differential output (connects to U1 electrode)
4	RF_OUT_N	Output (RF)	±1.8 V	High-speed negative differential output
5	VDD	Power	3.3 V	Power supply rail for the drain biasing circuit
6	VGG1	Input (DC)	0 to -2 V	Gate bias voltage tuning 1
7	VGG2	Input (DC)	0 to -2 V	Gate bias voltage tuning 2
8	DET_GND	Ground	0 V	Package central thermal ground pad connection (B.Cu vertical via-stitch)

8 Detailed Bill of Materials (BOM)

The complete manufacturing Bill of Materials (BOM) contains all the electrical and mechanical component specifications required for the Sseed Studio Fusion PCBA assembly line.

Table 9: Detailed Bill of Materials – Sseed Studio Fusion PCBA Template

Designator	Description	Manufacturer	Manufacturer Part Number
U1	TFLN Mach-Zehnder Modulator, 15mm, $V_{\pi} \leq 1.8\text{V}$, 100GHz	NTT / iXblue	TFLN-MZM-400G-C
U2	DFB Laser Diode, 1550nm, 100mW, C-band tunable	NeoPhotonics	TLN-1550-100
U3	High-Speed Photodetector, 100GHz, InGaAs PIN	Finisar / II-VI	XPDV4120R
U4	RF Driver IC, 100GHz, differential, 50 Ω , 3.3V	Analog Devices	HMC8410
U5	Laser Driver IC, 2.5Gbps, APC, < 100mA, SOP-8	Maxim Integrated	MAX3669
U6	Low-Noise LDO, 1.8V, 1A, 4.17 μVrms , PSRR 72dB	Texas Instruments	TPS7A4700
U7	Buck Converter, 3.3V, 4A, 2MHz, 95% efficiency, DFN-10	Analog Devices	LT8614
U8	TEC Controller, Peltier control, $\pm 5\text{A}$, 0.001°C stable	Wavelength	MPT5000
U9	SerDes IC, 400G PAM4, 100Gbaud retimer, FEC	Broadcom	BCM84881
U10	Clock Generator, 100GHz, < 50fs jitter, 12-Out	Silicon Labs	Si5395A
C1 - C20	MLCC Capacitor, 10 μF , 50V, X7R, $\pm 10\%$, SMD 1210	Murata	GRM32ER71H106KA12
C21 - C40	MLCC Capacitor, 0.1 μF , 25V, X7R, $\pm 10\%$, SMD 0603	Murata	GRM188R71E104KA01
C41 - C2095	MLCC Capacitor, 0.1 μF , 25V, X7R, $\pm 10\%$, SMD 0603	Murata	GRM188R71E104KA01
R1 - R10	Thin Film Resistor, 50 Ω , 0.1%, 0.1W, SMD 0603	Vishay	TNPW060350R0BEEN
L1 - L4	RF Inductor, 10 nH, $Q > 40$ @ 2GHz, SMD 0603	Coilcraft	0603CS-10NXJLW
J1	PCIe x16 Edge Connector, Gen5, 32GT/s, Gold plated	TE Connectivity	2-2013289-6
J2 - J3	LC/APC Fiber Connector, Single-Mode, APC polish	Senko	SN-LC-APC-SM
J4 - J7	SMA RF Connector, 50GHz, 50 Ω , edge launch	Amphenol	132289
PCB1	15-Layer Rogers RO4350B PCB substrate, IPC Class 3	Advanced Circuits	CUSTOM-15L-RF

9 Power Architecture Thermal Core Design

9.1 Regulator Tree and Low-Noise Rail Design

To preserve high signal integrity across the analog, RF, and optical modules, the LR-P3A power tree utilizes a high-efficiency cascading topology:

- **12V Main Input (J1 / 12VHPWR):** Accepts 12V DC power directly from the server. Prior to applying power, resistance measurements to GND must be checked to confirm isolation ($> 10\text{ k}\Omega$).
- **+3.3V Auxiliary Rail (U7 Buck Regulator):** Generated via the Analog Devices LT8614 buck converter. The converter switches at a high frequency of 2 MHz to push switching harmonics far outside the optical carrier band, delivering a 4A continuous output with 95% peak efficiency.
- **+1.8V Reference Rail (U6 LDO):** Linear regulator (Texas Instruments TPS7A4700) steps down power from the 3.3V rail. It supplies the photodiode monitors and laser bias circuits, operating with a very low noise floor ($4.17\text{ }\mu\text{Vrms}$) and high power supply rejection ratio (72 dB).
- **0.9V TFLN RF Rail (Fixed LDO):** Supplied by the ADP7118-0.9 low-dropout regulator to isolate the SerDes transmitter electrodes from high-frequency system power line noise.

9.2 CVD Synthetic Diamond Thermal Heat Core

The high density logic on the Gen3 NCE ASIC creates extreme local hotspots (tdp up to 800W per SoC with localized power densities exceeding 1000 W/cm^2). If lateral thermal pathing is used, heat accumulates rapidly, causing thermal throttling of the compute cores and wavelength drift on co-packaged optical waveguides.

To resolve this cooling bottleneck, an ultra-thin, **Chemical Vapor Deposition (CVD) synthetic diamond substrate layer** is grown directly underneath the silicon interposer.

- **Thermal Performance:** CVD diamond provides an isotropic thermal conductivity of **2,000 W/mK** (five times higher than copper).
- **Junction Control:** Acts as a highly efficient heat spreader, allowing heat to escape vertically into the liquid cold plate. This limits HBM4 stack junction temperatures to $\leq 95^\circ\text{C}$ and the logic core to $\leq 92^\circ\text{C}$ under an active 800W TDP load.

9.3 Peltier Thermoelectric Cooling (TEC) Loop

The optical wavelength response of TFLN Mach-Zehnder Modulators relies heavily on absolute physical temperature stability. The Wavelength Electronics MPT5000 TEC Controller (U8) manages internal Peltier cooling blocks to regulate localized waveguide zones at **$25^\circ\text{C} \pm 0.001^\circ\text{C}$** . This is crucial to prevent bias drift and preserve symmetric PAM4 eye openings under transient compute loads.

10 Manufacturing, Fabrication and Assembly Guidelines

The fabrication of the 15-Layer Rogers RO4350B PCB requires strict adherence to IPC Class 3/3A guidelines. The following manufacturing rules are compiled to guide the fabrication house during board synthesis:

10.1 Standard HDI Assembly Constraints

- **HDI Build Type:** HDI Type III Any-Layer construction. Minimum microvia drill size is set to **0.10 mm** with a copper-plated target annular ring of **0.05 mm** to accommodate high packaging densities.
- **Surface Treatment:** Electroless Nickel Immersion Gold (ENIG) coating per IPC-4552 Class 3 (Nickel thickness $\geq 3 \mu\text{m}$, Au $\geq 0.05 \mu\text{m}$ over $2 \mu\text{m}$ hard gold at the BGA pads) to support wire-bonding at the TFLN interposer facets.
- **Via Fill and Planarization:** All via-in-pad and micro-via arrays must be filled with non-conductive epoxy, copper-plated flat, and planarized to prevent void creation during solder reflow.

10.2 Back-Drilling on High-Speed Vias

Automated back-drilling is mandatory on all high-speed signal vias (SERDES_100G, PCIe_Gen6, TFLN_RF, and RF_50OHM *DIFFnetclasses*) *transitioning across layers*.

Stub Minimization: The residual back-drill stub must not exceed **0.127 mm (5 mils)** to eliminate parasitic reflections.

Tolerance: Back-drill depth accuracy must be held within $\pm 0.05 \text{ mm}$, verified with cross-section coupons post-lamination.

10.3 TFLN Keep-out and Solder Mask Specifications

- **Electromagnetic Keep-out:** A physical copper-free clearance zone of **100 mils (2.54 mm)** must be maintained on all internal and external copper layers surrounding the TFLN waveguide facets and the fiber-coupling arrays.
- **Solder Mask Expansion:** Minimum expansion of **2 mils (0.051 mm)** per side is enforced to prevent matte-black mask overlay on BGA or micro-optical alignment pads.

11 Production Test Plan Validation Protocols

The contract manufacturer must perform the following three-phase quality assurance testing sequence on every assembled board before system packing.

11.1 Phase 1: Unpowered Isolation Resistance Check

With the device fully unpowered, measure isolated path resistance to GND to check for solder bridges or component failure:

- **12V Input (J1 Pin B1):** Resistance to GND must measure $> 10 \text{ k}\Omega$.
- **3.3V Buck Rail (TP_3V3):** Resistance to GND must measure $> 500 \Omega$.
- **1.8V LDO Rail (TP_1V8):** Resistance to GND must measure $> 200 \Omega$.
- **V_CORE Supply (ASIC Rail):** Resistance to GND must measure $> 100 \Omega$.

11.2 Phase 2: Regulator Power-Up Sequencing

Apply current-limited 12V DC power (limit set to max 2A) to the primary edge pins. Measure DC voltage stability:

- Verify Buck Output (U7): Measure **3.3V** $\pm$ **2%** at test point TP_3V3.
- Verify LDO Output (U6): Measure **1.8V** $\pm$ **1%** at test point TP_1V8.
- Verify Reference Supply (U2/U5): Measure **2.5V** $\pm$ **5%** at test point TP_LASER_VCC.

11.3 Phase 3: High-Speed Optical Loopback Eye Testing

Connect the card to a PCIe Gen 6 compliance baseboard and run Fabric OS diagnostics:

- Trigger the NeoPhotonics DFB laser array (U2) and monitor back-facet photodiode current to ensure the output CW power matches 100 mW.
- Stream a 100 Gbaud PAM4 test sequence (PRBS31) through the TFLN modulators.
- Measure the resulting eye opening and error-rate at the photodetector output bank (U3). The receiver must achieve a native Bit Error Rate (BER) of $\leq 10^{-15}$ without Forward Error Correction (FEC).